Identifying OPP Labelled Nascent Proteome

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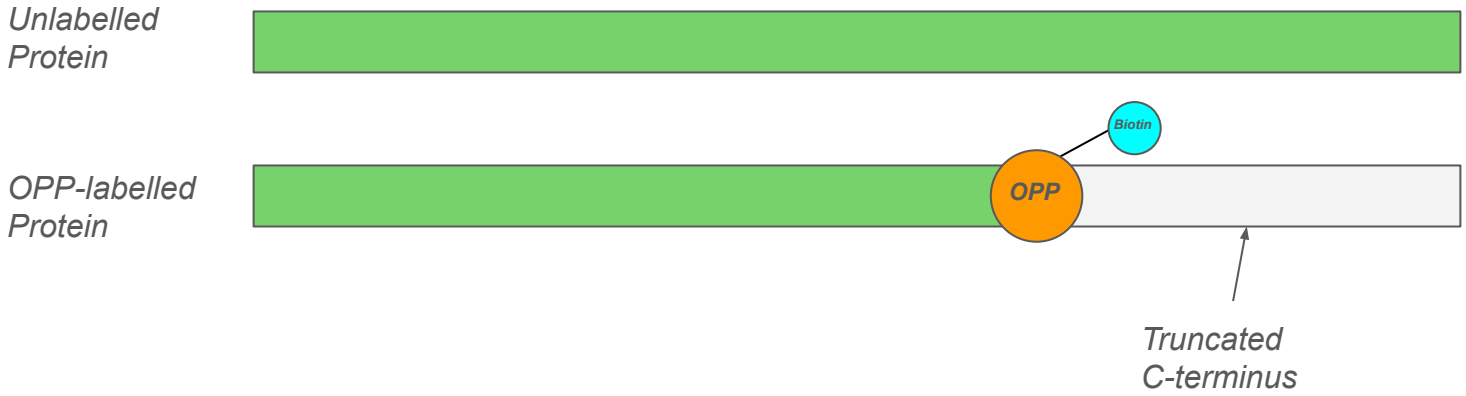
O-propargyl-purmomycin (OPP)-labelled proteomics data

- OPP incorporates into translating polypeptides
- Terminates translation → **enrichment at N-terminus expected**
- Challenge: identify labelled proteins from background data in labelled sample
- Dataset: Originally a degree project from 2021, continued with publicly availible dataset - PXD043498



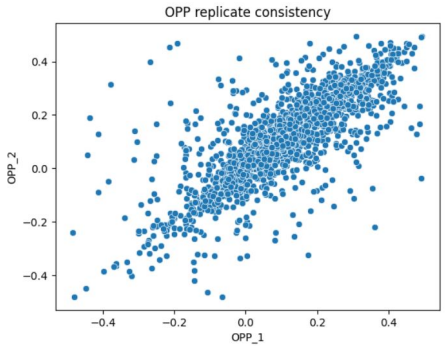
Method 1: Intensity Weighted Peptide Position Score

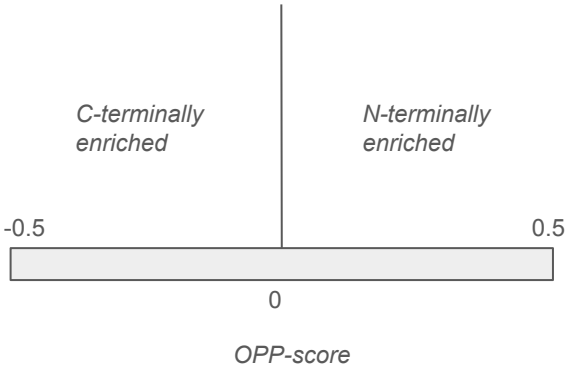
$$Score = 0.5 - \frac{\sum(I_i \cdot position_i)}{\sum I_i}$$

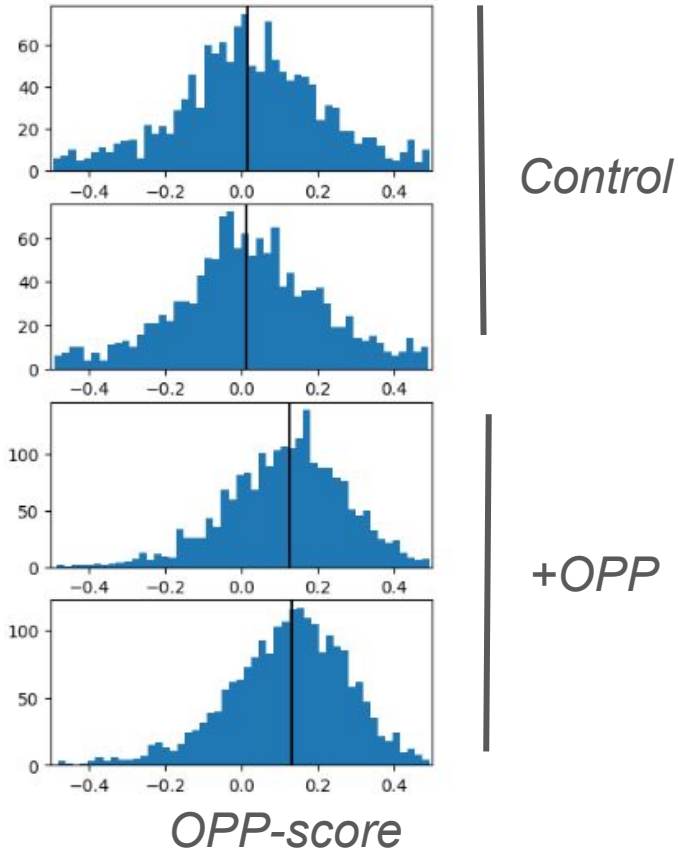
Higher score → C-terminal truncation

253 proteins with p-values < 0.05

None significant after FDR correction







Method 2: Calculating Significance at peptide level

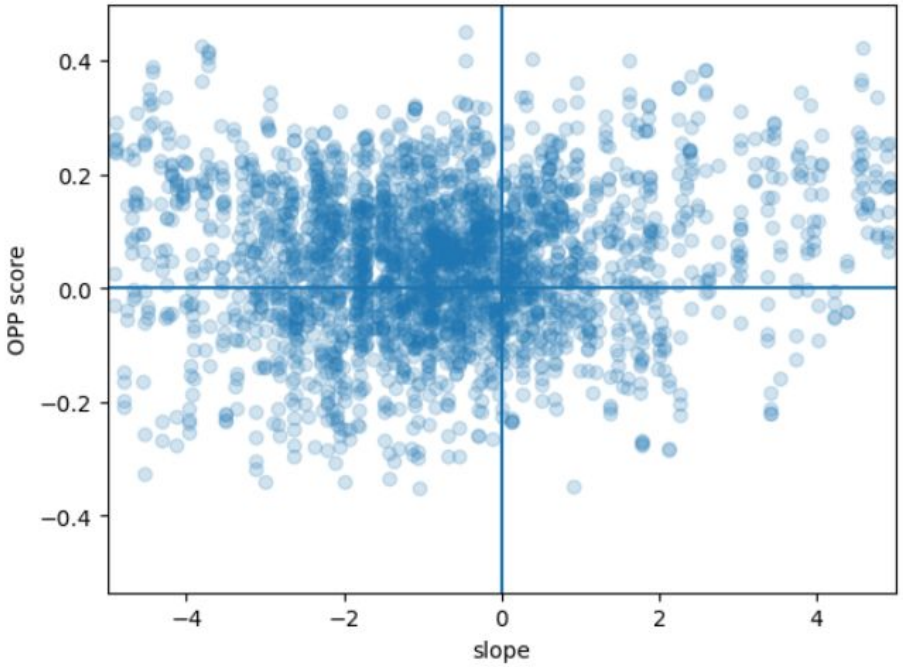
$$r_i = \log_2(I_{i,\mathrm{OPP}}) - \log_2(I_{i,\mathrm{DMSO}})$$

$$\sum_{i=1}^{n} w_i (r_i - \hat{r}_i)^2$$

$$r_i = \beta_0 + \beta_1 p_i$$

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$$

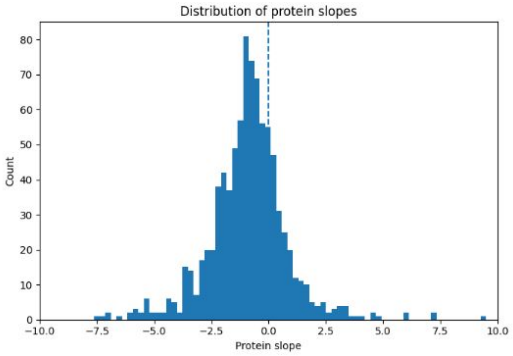
$$\mathrm{FDR} = \mathrm{BH}(p)$$

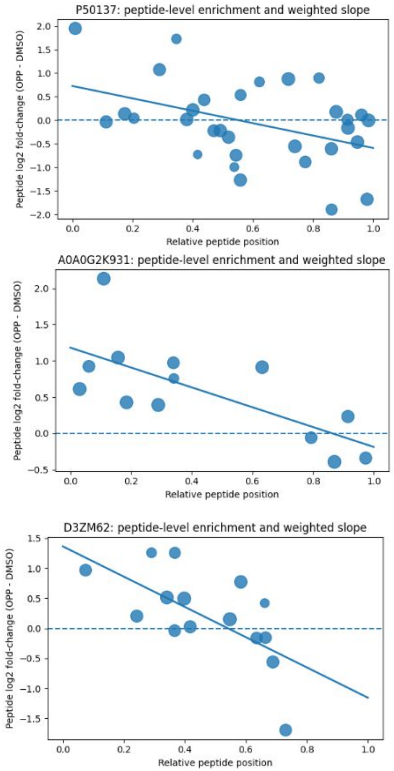


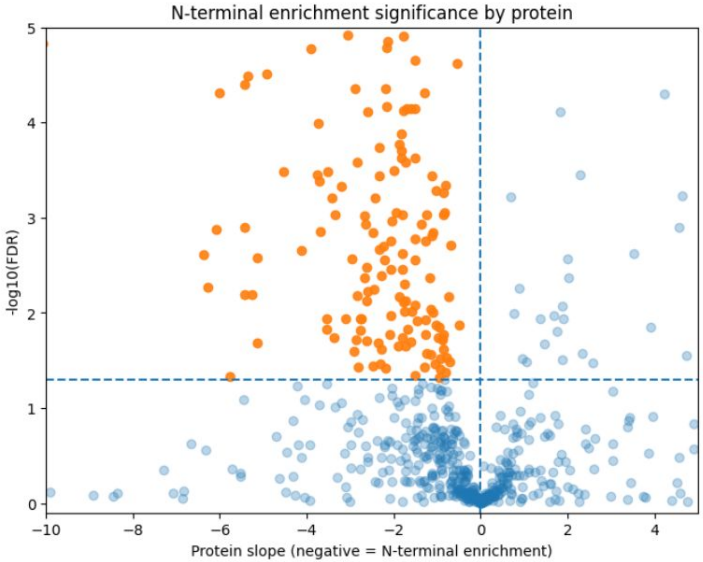
Negative slope = N-terminal enrichment

Calculating slope at peptide level reveals significantly N-terminally enriched peptides

- Identified 318 proteins with FDR < 0.05 negative slopes (tested 888 proteins)
- Captures N-terminal enrichment across protein.







Conclusions & Future Direction

Conclusions

- Slope-based metric identifies significantly N-terminally enriched proteins as expected with OPP labelling
- However, held back by missing peptide values due to stochastic nature of tandem mass spectrometry proteomics
- Many proteins excluded from analysis that are not found in control sample set

Future Direction

- For proteins identified in both sample sets with no matching peptides
  - Binning
- For proteins not identified in control set
  - Control experiment running through streptavidin pull down without OPP-labwelling
  - Similar proteins: coverage, no. peptides, total intensity (binning) - test for known proteins
- Use ratio of N-terminal / C -terminal ratio to avoid linear assumption
- Higher number of experimental repeats to make missing peptide values less likely
- Investigate significant proteins and their function.